

Industrial FM-to-C-band Spectrum-Sensing Node

SDR and Processing Backend Selection

The revised platform is a modular single-channel spectrum-sensing node for approximately 70 MHz to 6 GHz monitoring. Compared with the earlier USRP N210-based dual-branch concept, this version adopts an **industrial-ready, single-RX architecture** centered on a **USRP B200mini-i** and a **Jetson Orin Nano 8GB**. The change keeps full FM-to-C-band frequency coverage while reducing daughterboard cost, simplifying synchronization, and aligning the design with currently available hardware.

USRP B200mini-i Industrial SDR.

- The B200mini-i provides integrated RF coverage from **70 MHz to 6 GHz** through the AD9361 transceiver, avoiding the external daughterboard stack required by the legacy N210 solution. In this design it is used as a **single receive sensing front-end** with approximately **56 MHz instantaneous bandwidth, 12-bit conversion**, and USB 3.0 host connectivity.
- **Industrial deployment advantage:** the B200mini-i can be integrated into a ruggedized enclosure and is specified for **-40 °C to +75 °C** operation in the industrial configuration, which is a major advantage over used, non-industrial legacy SDRs.
- **Key trade-off versus the old N210 architecture:** this design gives up simultaneous dual-branch reception. Instead, the node time-shares a single RF path through antenna and filter switching. This reduces RF hardware cost and still supports wideband scanning missions.
- Estimated price: **1,100–1,400 USD** for the SDR plus industrial enclosure kit.
- **Open-source software support:** UHD remains the primary driver stack, with compatibility for GNU Radio, Python, C/C++, and SoapySDR-based workflows.

Jetson Orin Nano 8GB.

- The processing backend is upgraded from Jetson Nano 4GB to **Jetson Orin Nano 8GB**, reflecting current availability and a much stronger compute margin for field analytics, signal classification, and future edge-AI workflows.

- Key specifications: **6-core ARM Cortex-A78AE**, **8 GB LPDDR5**, Gigabit Ethernet, multiple USB 3.0 ports, and 40-pin GPIO for switch and attenuator control.
- Power envelope: approximately **15–25 W** depending on operating mode and workload. This is higher than the legacy Jetson Nano 4GB, but it enables substantially more on-node processing headroom.
- **System-level consequence:** because the B200mini-i is USB-only, the Orin Nano must be **co-located in the same enclosure** as the SDR. The Jetson then serves as the edge gateway, forwarding processed spectra and metadata over Ethernet to the network.
- Estimated price: **499–599 USD**.

Antenna System Configuration and RF Path Planning

Because the revised architecture uses a single sensing RX chain, the antenna and front-end planning changes from two simultaneous branches to one switched path:

*Selected antenna → protection → preselector bank → LNA/bypass →
programmable attenuation → USRP B200mini-i RX*

The earlier document separated VHF/UHF and UHF–C-band into parallel hardware branches. In the updated design, those bands remain covered, but they share one RF chain through solid-state switching. This is the main architectural shift introduced by the research context.

BicoLOG 5070 or equivalent VHF/UHF antenna → MSW-2-50DR+
antenna switch → Limiter / ESD / DC block
→ shared filter bank → LNA / bypass → ZX76-31R75PP-S+
attenuator → B200mini-i RX

HyperLOG 7060 or equivalent UHF–C-band antenna →
MSW-2-50DR+ antenna switch → Limiter / ESD / DC block
→ shared filter bank → LNA / bypass → ZX76-31R75PP-S+
attenuator → B200mini-i RX

Supporting Hardware

Prototype focus: single shared RF chain, switched antennas, shared filter bank, essential automation.

Field-ready focus: industrial enclosure, programmable attenuation, remote telemetry, thermal and power hardening.

Band	Antenna Type	Model Example	Frequency Range	Notes
VHF/UHF survey	Biconical	Aaronia 5070 BicoLOG	70 MHz–700 MHz	Preferred calibrated low-band antenna.
UHF–C-band survey	Log-periodic	Aaronia 7060 HyperLOG	700 MHz–6 GHz	Main upper-band survey antenna.
Low-cost alternative	Discone + log-periodic	Generic discone + HyperLOG	100 MHz–3 GHz plus 700 MHz–6 GHz	Lower cost, but reduced low-band confidence below 100 MHz.

Table 1: Antenna selection for the updated single-channel industrial node.

i) **Antenna switching and protection.**

The B200mini-i architecture benefits from a simple front-end antenna selection stage because only one receive path is active at a time. A practical baseline is a **Mini-Circuits MSW-2-50DR+** or equivalent **SPDT** RF switch between the low-band and high-band survey antennas. The switch is attractive because its **3.3V CMOS control** is compatible with Jetson GPIO, which keeps the control scheme close to the simpler GPIO-driven philosophy already present in the older design files.

Protection should remain the first RF element after the selected antenna. The research context recommends a limiter/ESD stage built around a gas-discharge arrestor, TVS protection, and DC blocking so the shared front-end is not exposed directly to outdoor transients.

ii) **Preselection and band plan.**

Although the B200mini-i tunes from 70 MHz to 6 GHz, the integrated transceiver does not remove the need for external selectivity. In the revised single-channel architecture, the filter bank becomes a **shared switched preselector** rather than one network per branch. This preserves the same monitoring regions discussed in the earlier draft, but with lower hardware duplication.

A deployable implementation can realize this bank using a compact **SP6T** or a cascaded solid-state switch arrangement built from parts in the same class as the **MSW-2-50DR+**. The earlier prototype idea of manual patch-cord routing remains acceptable for bench validation, but the research-backed field version assumes automated selection.

iii) **LNA and bypass strategy.**

The gain plan remains frequency dependent, but it is now centralized into one shared chain. This reduces the number of active RF modules relative to the old dual-branch draft.

The LNA stage should be placed after the selected preselector path, not before it, so strong out-of-band signals are attenuated prior to gain.

Sensing Region	Re-	Suggested Path	Filter	Nominal Band	Typ. IL	Purpose and Comments
70–88 MHz		Low-VHF path or broad VHF preselector		70–88 MHz	TBD	Supports lower-VHF sensing beneath the FM broadcast band.
88–108 MHz		Switchable FM notch or FM pass		88–108 MHz	<1.5 dB pass	Protects the receiver during surveys, but can be bypassed when FM is the target band.
108–700 MHz		Mini-Circuits SHP-50+ + SLP-750+		108–700 MHz	~2 dB	General VHF/UHF preselection before amplification.
700–960 MHz		Modular BPF or cavity filter		700–960 MHz	<2 dB	Useful for upper-UHF, public-safety, and telemetry regions.
1.7–2.7 GHz		Custom BPF or VBFZ-2130-S+ style path		1.7–2.7 GHz	<2 dB	Targets cellular, ISM, and related microwave services.
3.0–4.2 GHz		Mini-Circuits SBP-3800+		3.0–4.2 GHz	<3 dB	Concrete catalog part retained from the research note to close the 3–4 GHz gap.
4.0–6.0 GHz		Marki Microwave C-band BPF or equivalent		4.0–6.0 GHz	<3 dB	Replaceable upper-band module for focused C-band sensing.

Table 2: Shared RF band plan for the B200mini-i single-channel sensing node.

Model	Range	Gain	NF	Use Case
Mini-Circuits ZX60-33LN-S+	50 MHz–3 GHz	20 dB	3 dB	Main LNA path for VHF, UHF, and lower microwave sensing.
Optional 3–4 GHz LNA	3.0–4.2 GHz	15–25 dB	TBD	Optional gain flattening path when the 3–4 GHz region is mission-critical.
Amplitech AGLNA035082	4.0 GHz–8.0 GHz	26 dB typ.	0.9 dB	C-band and upper microwave amplification path.
Bypass path	DC–6 GHz class	0 dB	–	Allows protected operation near strong emitters and prevents unnecessary compression.

Table 3: Frequency-dependent gain options for the shared RF chain.

In a compact deployable version, the Jetson GPIO can command the LNA/bypass state in the same way that the older design used GPIO for RF switching.

iv) Attenuation strategy.

The research context strongly favors a programmable attenuator for field use, because a single unattended node must adapt to both quiet rural environments and high-power urban RF scenes. The selected part is:

- **Mini-Circuits ZX76-31R75PP-S+**: **0–31.5 dB** in **0.5 dB steps**, DC–6 GHz class. Estimated: **300–350 USD**.

This is a meaningful departure from the earlier prototype approach that relied on manual fixed SMA pads. The fixed-pad idea is still valid

for laboratory prototyping, but the new document assumes a more deployable baseline where remote gain control is part of the default hardware stack.

v) **Timing and synchronization.**

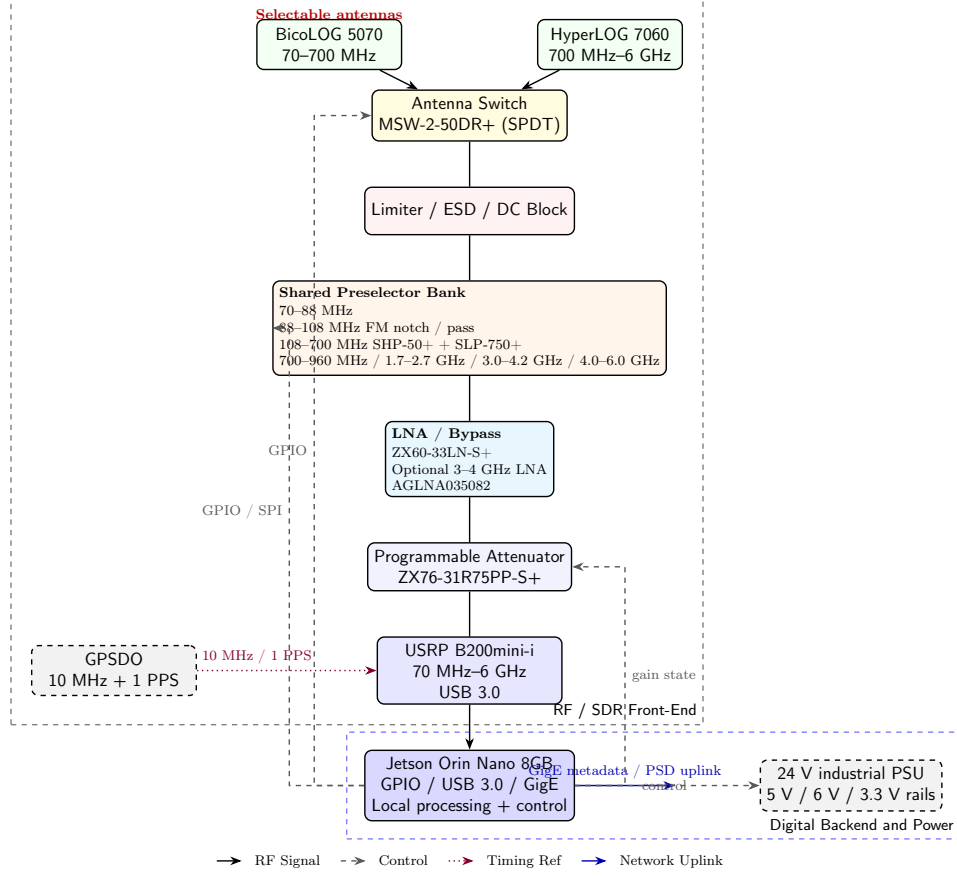
The B200mini-i can operate from its onboard reference for many scanning tasks, but the updated architecture still benefits from an external timing option when metadata consistency or multi-node comparison matters.

- **External reference input:** 10 MHz and 1 PPS via SMA.
- **Suggested source:** Leo Bodnar GPSDO or equivalent compact GPSDO module. Estimated: **50–150 USD**.
- The node software should log reference-lock and GPS-lock state alongside capture metadata.

System Block Diagram

The following diagram summarizes the revised single-channel industrial node. It keeps the component-selection flavor of the original document, but reflects the research-backed B200mini-i + Orin Nano architecture.

B200mini-i + Jetson Orin Nano 8GB Architecture



Total Budget Assessment of the Bill of Materials

The revised bill of materials follows the industrial single-channel architecture from the research context. The biggest cost changes relative to the older draft are the removal of N210 daughterboards and duplicated RF branches, and the addition of ruggedized power and enclosure elements.

Key budget interpretation.

1. **Savings driver:** eliminating the N210 daughterboard stack and the second RF branch reduces duplicated filter, LNA, and attenuation costs.
2. **Industrial cost driver:** the enclosure, power conversion, and thermal management budget is more important than in a lab-only build because the USB-only SDR and the Jetson must live in the same enclosure.
3. **Automation assumption:** unlike the older draft, this budget assumes programmable attenuation and switched RF control are normal parts of the deployable configuration.

Subsystem	Estimated Cost	Comments
USRP B200mini-i with industrial enclosure	1,100–1,400 USD	Integrated 70 MHz–6 GHz SDR. Removes daughterboard cost and legacy used-market risk.
Jetson Orin Nano 8GB	499–599 USD	Main local controller, USB host, GPIO source, and edge-processing backend.
Antenna system	580–730 USD	BicoLOG 5070 plus HyperLOG 7060. Lower-cost alternatives exist but reduce calibration confidence.
Antenna SPDT switch	25–40 USD	MSW-2-50DR+ or equivalent. Used for low-band versus high-band antenna selection.
Limiter / ESD / lightning protection	40–80 USD	Shared protection stage for the single RF chain.
Preselector filter bank	500–650 USD	Includes FM handling, VHF/UHF preselection, upper-UHF, 1.7–2.7 GHz, SBP-3800+, and C-band path allowance.
LNA paths and bypass	220–350 USD	ZX60-33LN-S+, AGLNA035082, bypass hardware, and optional 3–4 GHz gap path.
Programmable attenuator	300–350 USD	ZX76-31R75PP-S+ as default deployable attenuation strategy.
GPSDO timing reference	50–150 USD	Compact external timing module for 10 MHz and 1 PPS.
RF cables, adapters, terminators	100–200 USD	Low-loss SMA interconnects and integration accessories.
IP65 enclosure, thermal management, mounts	150–300 USD	Industrial packaging for co-located USB-only SDR and Jetson.
24 V PSU and isolated DC-DC rails	60–120 USD	Supports separate 5 V, 6 V, and 3.3 V domains.
Calibration and test allowance	100–250 USD	Reference loads, verification accessories, and setup margin.
Estimated subtotal (hardware)	3,724–5,219 USD	Aligned with the industrial-ready single-channel architecture in the research note.
Contingency, about 15%	560–780 USD	Covers sourcing variation, enclosure details, and filter-path adjustments.
Defensible project budget	4,300–6,000 USD	Recommended planning range for a field-deployable industrial sensing node.

Table 4: Budget assessment for the revised B200mini-i + Orin Nano spectrum-sensing node.

Power Budget Assessment

The power budget is no longer dominated by a low-power Jetson Nano prototype. The Orin Nano and the industrial support hardware raise the total requirement, but the design remains modest enough for a small industrial DC supply.

Subsystem	Supply Rail	Typical Power	Maximum Power	Power-Budget Notes
Antennas and passive filters	–	0 W	0 W	Passive RF elements.
Antenna and filter switching	5 V control	0.2–0.6 W	1.0 W	Solid-state selection hardware for antenna and shared preselector bank.
ZX60-33LN-S+ path	5 V RF rail	0.3–0.5 W	0.7 W	Main low-to-mid-band amplification path.
AGLNA035082 path	5 V RF rail	0.3–0.7 W	1.0 W	Upper-band LNA path.
Programmable attenuator	5 V control	<0.1 W	0.2 W	Low DC burden, but operationally important.
GPSDO timing module	5 V rail	0.5–1.0 W	1.5 W	Optional but recommended for disciplined timing.
USRP B200mini-i	6 V DC / USB	2.0–4.0 W	6.0 W	USB-connected SDR located inside the same enclosure as the Jetson.
Jetson Orin Nano 8GB	5 V system rail	10–20 W	25 W	Main processing backend in 15 W or higher operating modes.
Storage, cooling, GPIO, auxiliaries	5 V aux rail	1.0–3.0 W	5.0 W	NVMe or microSD, fans, logic support, interface overhead.
Estimated system total	–	14.6–30.3 W	40.4 W	Expected envelope for the industrial single-channel node.
Recommended supply capacity	–	20–35 W	50 W	Allows startup margin, temperature effects, and future peripherals.

Table 5: Estimated power budget for the revised industrial sensing node.

A practical deployment should use a **24 V DC industrial supply** feeding isolated conversion stages for the digital and RF rails. This mirrors the research note and is more realistic than a bench-style single-adaptor approach.

Suggested supply-rail organization.

Rail	Recommended Rating	Loads	Notes
Main digital rail	5 V, 6 A	Jetson Orin Nano, storage, cooling, auxiliary logic	Sized for sustained Orin operation plus peripherals.
SDR rail	6 V, 3 A	USRP B200mini-i	Kept separate from the main digital rail for cleaner integration.
Low-noise RF rail	5 V, 2–3 A	LNAs, GPSDO, RF switching, attenuator control	Derived through low-noise regulation.
Control rail	3.3 V logic	GPIO-compatible control paths	Used for switch logic and related digital control.

Table 6: Recommended supply organization for the B200mini-i industrial node.

Deployment Notes and Design Risks

1. **USB 3.0 co-location requirement:** unlike Ethernet-based N210 deployments, the B200mini-i must stay within a short USB 3.0 link distance of the Jetson. The enclosure therefore becomes a first-class design element.
2. **Single-channel limitation:** the node cannot monitor two bands simultaneously. If that becomes mission-critical, duplicating the full node is a clearer path than forcing a return to the more expensive dual-branch design.
3. **3–4 GHz flatness risk:** if this region is a key sensing target, the optional gap LNA should be treated as a likely requirement rather than a nice-to-have.
4. **Thermal design:** the Orin Nano power density and the enclosed SDR mean airflow and heat conduction must be validated early.
5. **Calibration discipline:** repeatable field measurements still depend on characterizing the full chain from antenna through attenuation and SDR input.

Conclusion

The updated initial design now converges on a more defensible industrial sensing node: **USRP B200mini-i** for integrated 70 MHz–6 GHz SDR coverage, **Jetson Orin Nano 8GB** for local control and future edge analytics, a **shared switched preselector chain** instead of duplicated RF branches, and concrete front-end parts such as **MSW-2-50DR+**, **SBP-3800+**, **ZX60-33LN-S+**, **AGLNA035082**, and **ZX76-31R75PP-S+**.

Relative to the older documents, the biggest conceptual change is not the frequency coverage but the system packaging: one industrial SDR, one stronger embedded backend, one automated RF path, and a co-located enclosure designed for remote field operation.